Fourier Series Th$^{\underline{m}}$

any function $f(x)$ defined on interval $(0,a)$

can be written as a sum of sine and cosine waves

$e^{i(kx-\omega t)}$

$e^{ik\cdot r}$

$$f(x) = \sum_{n=\infty}^{\infty} C_n e^{-ik_n x}$$

↑ weight factor

$$a_n = \frac{1}{A}\int_{-A}^{A} f(x)\cos\frac{n\pi x}{A}\,dx$$

where $C_n = \frac{1}{A}\int_0^a f(x)e^{ik_n x}\,dx$

$$b_n = \frac{1}{A}\int_{-A}^{A} f(x)\sin\frac{n\pi x}{A}\,dx$$

$$k_n = n\left(\frac{2\pi}{a}\right) \qquad (c_0, c_1, c_2, \ldots)$$

Fourier Transform Th$^{\underline{m}}$

* each contains all of the information of the wave function:

'different representations' of same function

can convert between kspace / real space freely

real space function

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(k)e^{-ikx}\,dk$$

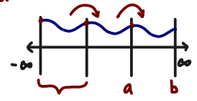
↑ weight factor

k-space function

$$F(k) = \int_{-\infty}^{\infty} f(x)e^{ikx}\,dx$$

for all 'well behaved' $f(x)$

↳ 'if it doesn't have a pole'



periodic function

$f(x+L) = f(x)$

↑

period = boundary condition

→ finite of k's

* Fourier series - special case of Fourier Transform for a periodic function

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*conceptual: explain the Born Rule in terms of probability

Special Case: "Gaussian wave packet"

normal "distribution"

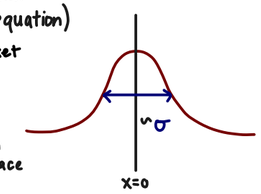
(possible solution of Schrödinger equation)

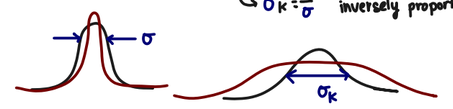
$f(x) = e^{-x^2/2\sigma^2}$ ← width of packet

$$f(x) = \sqrt{2\pi\sigma^2}\,\underbrace{e^{-k\sigma^2/2}}_{e^{-k/2\sigma_k^2}}$$

← width in k-space

↳ $\sigma_k = \frac{1}{\sigma}$ inversely proportional





$f(x) = e^{-ik_0 x}$

$F(k) = 2\pi\,\delta(k-k_0)$

calculate $F(x)$: $F(k) = \int_{-\infty}^{\infty} dx\, \underbrace{e^{-ik_0 x}}_{f(x)}\, dx\, e^{ikx}\,dx$

$$= \int_{-\infty}^{\infty} dx\, e^{i(k-k_0)x}$$

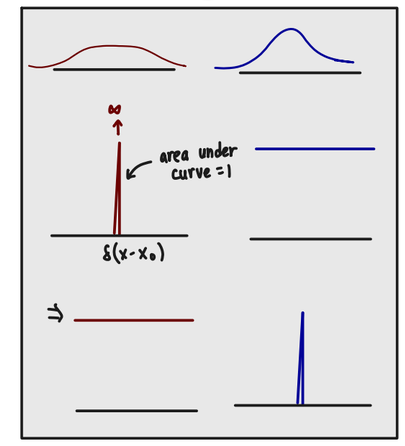
$e^{i\theta} = \cos\theta + i\sin\theta$ $\qquad e^{i(0)} = 1$

if $k \neq k_0 \to 0$

if $k = k_0 \to \infty$

real space (position) vs. k-space (momentum)

small in real ⇄ large in k



⇒ General Property of δ-function: $\int_{k_1}^{k_2} f(k)\,\delta(k-k_0)\,dk = f(k_0)$ if $k_1 < k_2 < k_3$

how do we normalize a plane wave?

$$\int_{-\infty}^{\infty} \psi^*(x)\,\psi(x)\,dx = 1$$

↓

$$\lim_{L\to\infty}\int_{-L/2}^{L/2} (Ae^{\cancel{ik_0 x}})(Ae^{\cancel{-ik_0 x}})\,dx = L\cdot A^2 = 1$$

→ $A = \frac{1}{\sqrt{L}}$

→ $\psi(x) = \frac{1}{\sqrt{L}}e^{-ik_0 x}$

3D: $\psi(\vec{x}) = \frac{1}{\sqrt{V}}e^{-i\vec{k}_0\cdot\vec{x}}$

$|\psi(x)|^2 = \frac{1}{V}$

Measurement Probabilities.

Born Rule: probability of finding particle between a and b

$$= \int_a^b |\psi(x)|^2\,dx$$

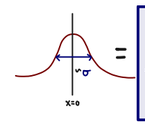
for other measurements, do "weighted average" or expectation value

$$\left.\begin{aligned} \langle x\rangle &= \int dx\, \psi^*(x)\, x\, \psi(x) \\ \underset{mv}{\langle p\rangle} &= \int dx\, \psi^*(x)\underbrace{\left(-i\hbar\frac{d}{dx}\right)}_{p\text{ 'operator'}}\psi(x) \end{aligned}\right\} = \int dk\, \psi^*(k)\underbrace{(\hbar k)}_{p}\underbrace{\psi(k)}_{\text{fourier transform}}$$

Probability example with Gaussian Wave Packet:

$$\langle x\rangle = \int dx \underbrace{\frac{e^{-x^2/2\sigma}}{(\pi\sigma^2)^{1/4}}}_{4r} \times \underbrace{\frac{e^{-x^2/2\sigma}}{(\pi\sigma^2)^{1/4}}}_{4r}$$

← normalization factor: $\int_{-\infty}^{\infty}\psi^*(x)\,\psi(x) = 1$



$$= \boxed{\int_{-\infty}^{\infty} dx\, \frac{e^{-x^2/2\sigma^2}}{(\pi\sigma^2)^{1/2}}\,x^2} = \frac{\sigma^2}{2}$$

same for k-space:

→ "uncertainty" $\Delta x = \frac{\sigma}{\sqrt{2}}$, "uncertainty" $\Delta k \to \frac{\sigma}{\sqrt{2}}k = \frac{1}{\sqrt{2}}\sigma$

multiply $\Delta x\Delta k = \frac{1}{2}$

$p = \hbar k = mv$

"incompatible measurements"

$E > \hbar\omega$

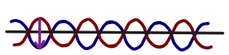
$F(\omega)F(t)$ $\qquad \Delta E\Delta t$ → virtual particle pairs

arise from 'natural properties' of Fourier Transform

Dirac Equation: Relativity + Quantum Mechanics

In general, we want a Relativistic model of QM for fermions.

recall,

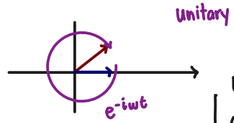


plane of light has $E(\bar{x},t) = 0$ everywhere @ some t

→ Dirac wanted unitary dynamics

linear derivative in time:

$E = \hbar\omega \to \frac{i\hbar\, d\psi}{dt} \to i\hbar\omega$



unitary evolution

Relativistic Invariant:

$E^2 = (mc^2)^2 + (pc)^2$

↳ $(E^2 + \Delta^2) = 0$

↳ $(E+\Delta)(E-\Delta) = 0$